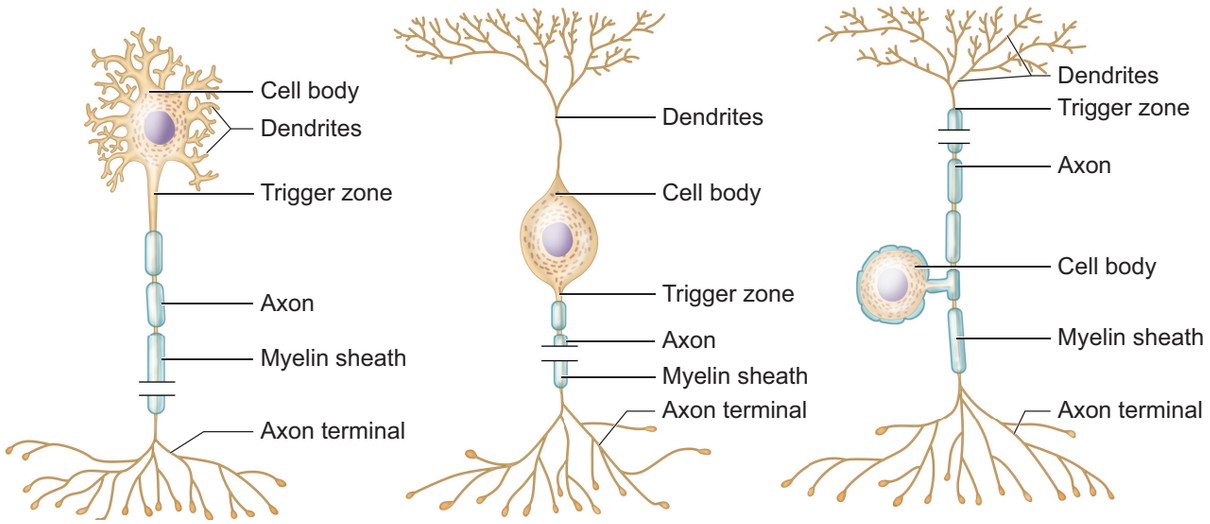


Figure 10.2 Types of Neurons

10.3.1 Generation and conduction of nerve impulses

This section deals with how the nerve impulses are produced and conducted in our body. Sensation felt in the sensory organs are carried by the nerve fibres in the form of electrical impulses. A nerve impulse is a series of electrical impulses, which travel along the nerve fibre. Inner to the axolemma, the cytoplasm contains the **intracellular fluid** (**ICF**) with large amounts of potassium and magnesium phosphate along with negatively charged proteins and other organic molecules.**The extra cellular fluid (ECF)** found outside the axolemma contains large amounts of sodium chloride, bicarbonates, nutrients and oxygen for the cell; and carbon dioxide and metabolic wastes released by the neuronal cells. The ECF and ICF (cytosol) contains negatively charged particles (anions) and positively charged particles (cations). These charged particles are involved in the conduction of impulses.

The neurons maintain an uneven distribution of various inorganic ions across their axolemma for transmission of impulses. This unequal distribution of ions establishes the membrane potential across the axolemma. The axolemma contains a variety of membrane proteins that act as ionic channels and regulates the movement of ions across the axolemma. (Shown in Table 10.1).

10.3.2 Transmission of impulses

The transmission of impulse involves two main phases; **Resting membrane potential** and **Action membrane potential.**

Resting membrane Potential: The electrical potential difference across the plasma membrane of a resting neuron is called the **resting potential** during which the interior of the cell is negative due to greater efflux of K^+ outside the cell than

Note: The charged particles have potential energy. The potential difference is the measure of potential energy between two points which is measured in volts or millivolts.

TN_GOVT_XI_Zoology_Ch10.indd 179 09-01-2020 5.18.55 PM